

Are Moral Dilemmas Merely Technical Problems?

Executive summary

The thesis claims that (i) all moral dilemmas are “merely technical problems,” (ii) any ethical problem has a technical solution *in principle* if science advances far enough, and (iii) while new ethical problems may arise, newer/better technology can solve those as well. This is an exceptionally strong form of technological solutionism, combining a **reduction claim** (“moral $\rightarrow$ technical”) with an **unbounded progress claim** (“sufficient science $\rightarrow$ solution”), plus a **self-repair claim** (“technology’s externalities are themselves technologically solvable”). Hardin’s classic formulation of “no technical solution problems” is a direct, textually explicit denial of exactly this package: he argues that certain problems “require a fundamental extension in morality,” and defines a “technical solution” as one requiring changes in technique but “little or nothing” in values or morality. ¹

A careful philosophical and empirical assessment suggests a mixed conclusion. Technology often *does* dissolve ethical dilemmas by changing the feasible set (new options, reduced scarcity, better prediction, safer systems). Weinberg explicitly argues that “social problems” can sometimes be “circumvented” by reducing them to “technological problems,” seeking “Quick Technological Fixes” that do not require changes in “social attitudes.” ² Multiple case studies (vaccination, antibiotics, safer infrastructure, certain medical breakthroughs) show dramatic reductions in morally urgent harms. For example, WHO estimates immunization prevents **3.5–5 million deaths per year**. ³

However, the “all” and “always” in the thesis are vulnerable to several principled counterarguments: (a) some dilemmas are **normative conflicts among incommensurable values** (pluralism), not merely gaps in technique or information; (b) genuine dilemmas can leave **moral remainder/residue** even after the best action (Williams: “not all soluble without remainder”); ⁴ and (c) technology can embed and amplify value choices rather than decide them, raising meta-questions about legitimacy, authority, and responsibility (Habermas’s contrast between instrumental action governed by technical rules and democratic legitimacy; ⁵ Jonas’s argument that modern technological power changes the “nature of human action” and “calls for a change in ethics”; ⁶ Heidegger’s warning that treating technology as neutral blinds us to its “essence.” ⁷)

Empirically, powerful technologies routinely **create** ethical problems whose “solutions” are not purely technical (nuclear deterrence vs catastrophic risk; algorithmic governance vs rights and legitimacy; genome editing vs human dignity and consent across generations; climate intervention vs global governance). Bostrom emphasizes that technological revolutions “reshape culture and even—perhaps—human nature,” complicating any claim that progress yields straightforward “solutions.” ⁸ The most defensible “pro-thesis” position is therefore a *weakened* variant: many moral dilemmas have substantial technical components; science can often expand options; and technology can implement ethically chosen policies—but **technology does not by itself determine which ends are justified**, nor eliminate all forms of moral conflict and remainder.

Thesis, scope, and key definitions

Thesis under review (explicitly):

"All moral dilemmas are merely technical problems; there is always a technical solution if science advances enough to any given ethical problem; new ethical problems may arise but newer/better technology can solve them."

Scope and framing: This report evaluates the thesis as a claim about (1) the **metaphysics/logic of moral dilemmas** (are any "genuine" dilemmas irreducible?), (2) the **epistemology of moral decision-making** (is improved knowledge/forecasting sufficient?), (3) the **philosophy of technology** (is technology neutral instrument or world-shaping practice?), and (4) **empirical historical evidence**.

Definitions of key terms

To keep the thesis testable, terms must be defined at a level that separates *technical feasibility* from *normative justification*.

Moral dilemma (strict sense): A situation in which an agent faces conflicting moral requirements such that not all can be satisfied. Reference works emphasize the central dispute: whether "genuine" (ontological) dilemmas exist or whether dilemmas are merely apparent/epistemic. ⁹

Ethical problem: A broader class than "moral dilemma": any situation requiring moral evaluation, justification, or policy design under normative standards. Moral dilemmas are a subset of ethical problems (those with conflict among obligations/reasons). ¹⁰

Technical problem: A problem whose solution can be specified and validated primarily by **instrumental reasoning** (means-end optimization) under agreed goals and constraints. Hardin offers a useful operational contrast by defining a "technical solution" as one requiring "change only in the techniques" of natural science and "little or nothing" in values or morality. ¹

Technological solution: An intervention in material, biological, or informational systems (engineering, tools, infrastructure, algorithms) that changes feasible actions, constraints, or outcomes. Heidegger notes that the common "instrumental and anthropological definition" understands technology as "a means to an end" and "a human activity," while warning this can miss the "essence of technology." ⁷

Important ambiguity (load-bearing): The thesis slides between two senses: 1. **Weak/implementation sense:** technology can help implement a morally chosen policy (e.g., safer cars reduce fatalities), versus 2. **Strong/decision sense:** technology can determine what is morally right (resolve normative disagreement).

Hardin's definition highlights that much of what people call "technical solution" explicitly excludes changes in values. ¹ The thesis appears to require the *strong* sense to be true "always," but many philosophical objections target precisely that leap.

Philosophical landscape for and against the thesis

This section maps major historical and contemporary positions relevant to technological optimism/solutionism. The table also marks which figures are “**unspecified**” with respect to endorsing the thesis (they may diagnose technology’s role without claiming “all moral dilemmas are technical”).

Comparative table of philosophers and positions

Thinker (examples requested)	Primary orientation	What they plausibly support/reject about the thesis	Short primary-source quote (≤25 words)
Francis Bacon	Programmatic optimism about knowledge as power over nature	Supports a strong link between science and human power; can be read as precursor to solutionism, though not a modern ethics theory	“Human knowledge and human power meet in one.” ¹¹
Auguste Comte	Positivism; science as basis of social order and progress	Supports scientific optimism: social life governed by laws; knowledge enables prediction/action; risks technocracy	“From Science come Prevision; from Prevision comes Action.” (quoted as Comte’s “own words.”) ¹²
Alvin Weinberg	Explicit “technological fix” advocate	Directly supports reducing social problems to technical ones; seeks “Quick Technological Fixes” that avoid attitude change	Asks about “Quick Technological Fixes... within the grasp of modern technology.” ²
Martin Heidegger	Critique of instrumental view; technology as “enframing”	Rejects “merely technical” framing; argues neutrality/instrumentality is insufficient and can blind us	“The essence of technology is by no means anything technological.” ⁷
Hannah Arendt	Modernity critique; “world alienation”; space/science	Rejects naive solutionism; emphasizes losses in public world and political action linked to modern trajectories	“The purpose... is to trace back modern world alienation... its twofold flight...” ¹³
Jürgen Habermas	Distinguishes instrumental vs communicative rationality	Rejects reducing legitimacy/values to technique; stresses that instrumental action is rule-governed but not value-justifying	“Instrumental action is governed by technical rules...” ⁵

Thinker (examples requested)	Primary orientation	What they plausibly support/ reject about the thesis	Short primary-source quote (≤25 words)
Hans Jonas	Ethics for technological age; responsibility to future	Rejects “technology solves ethics” framing; argues tech changes action and thus demands new ethics and responsibility	“The nature of human action has changed... calls for a change in ethics.” ⁶
Michael Sandel	Critique of enhancement; humility, “giftedness”	Rejects mastery-as-solution; argues tech can corrupt moral relations (parenting, achievement)	“Openness to the unbidden” (via William F. May) and critique of “drive to master the mystery of birth.” ¹⁴
Langdon Winner	Technology is political; “forms of life”; critique of progress myth	Rejects neutrality and pure technical framing; stresses technologies shape social order and we “sleepwalk” through this	“We so willingly sleepwalk through the process of reconstituting the conditions of human existence.” ¹⁵
Bruno Latour	Actor-network theory; dissolves society/technology divide	Unspecified on “always solvable,” but supports idea that morality can be delegated/inscribed into artifacts and networks	Metal weight succeeds “where... moral obligation... failed.” (hotel key example) ¹⁶
Nick Bostrom	Tech as transformative; optimistic about enhancement; strong risk awareness	Mixed/conditional: technology can vastly expand capability, but risks and value alignment problems are central	“Superintelligence would be the last invention... biological man would ever need to make.” ¹⁷

Two conceptual lineages

A useful way to see the debate is as a clash between **instrumental-progress narratives** and **critical/constitutive narratives**.

1. **Instrumental-progress narrative (proto-solutionism):** Bacon’s science-as-power framing ¹¹ and Comtean positivism ¹⁸ provide a vocabulary in which social problems look increasingly law-governed and therefore controllable. Weinberg turns this into explicit policy advice: reduce social issues to engineering when possible and bypass attitude change. ²
2. **Critical/constitutive narrative:** Heidegger argues that treating technology as neutral instrument is a form of blindness; the “essence” of modern technology (enframing) reorganizes how reality appears and is valued. ⁷ Winner and Latour, in different idioms, claim technologies are not merely tools but *forms of life* or *networks of humans and nonhumans* that stabilize norms and power. ¹⁹ Jonas argues that modern power changes the moral situation itself: ethics must expand. ⁶ Arendt

locates a “world alienation” in modern trajectories of science and society. ²⁰ Habermas insists that legitimacy and values cannot be replaced by technical rules alone. ⁵ Sandel critiques enhancement as hubristic mastery that distorts moral relations. ¹⁴

Argument analysis of the thesis

Formal reconstruction

Let:

- **MD(x)** = x is a moral dilemma
- **TP(x)** = x is a technical problem
- **TS(x)** = a technological solution exists for x
- **SA(t)** = science advances sufficiently at time t

The thesis can be formalized as:

1. **Reduction:** $\forall x (MD(x) \rightarrow TP(x))$
2. **Sufficiency:** $\forall x (EthicalProblem(x) \rightarrow (\exists t SA(t) \rightarrow TS(x)))$
3. **Recursive solvability:** $\forall y (TechCreates(y) \rightarrow \exists t' SA(t') \rightarrow TS(y))$

This package implies a strong metaethical and practical claim: moral difficulty is always ultimately a matter of **constraints, information, and design**.

Where the thesis is strongest

The thesis is strongest when moral dilemmas arise from **constraint-induced collisions** that technology can relax:

- **Scarcity:** If two duties conflict because resources are insufficient, innovation can create additional options (e.g., better treatments, synthetic substitutes).
- **Coordination and incentive design:** If the dilemma reflects collective action traps, technology can reduce monitoring costs, increase transparency, or change the payoff structure—Latour’s “delegation” example shows how a material artifact can accomplish compliance where “moral obligation” failed. ¹⁶
- **Risk reduction:** If the dilemma depends on uncertainty about outcomes, science can improve prediction (but note: better prediction does not itself determine values).

Weinberg’s paradigm is precisely this: “circumvent” social problems by reducing them to engineering problems and implementing a “Quick Technological Fix” to avoid changing attitudes. ² In this sense, technology can sometimes convert a moral dilemma into a more routine design choice among non-catastrophic options.

Core vulnerabilities

The thesis fails if **any** of the following are true.

Non-reducibility of normative conflict (value pluralism / incommensurability):

Some ethical problems are conflicts among legitimate values that are not reducible to a single commensurable metric (e.g., liberty vs equality, privacy vs security). Even Hardin—while arguing some “incommensurables” become commensurable with weighting—still frames the need for a “theory of weighting,” i.e., a normative choice not delivered by technique alone. ¹ Berlin’s value pluralism tradition emphasizes conflicting, sometimes incommensurable values; scholarship on Berlin explicitly associates pluralism with difficulty of prioritizing values. ²¹

Moral remainder / moral residue:

Even when one acts rightly (or “for the best”), there can be leftover moral cost. Williams states bluntly: “Moral conflicts are neither systematically avoidable, nor all soluble without remainder.” ⁴ If remainder is a constitutive feature of moral conflict, then “technical solution” (in the sense of fully resolving the dilemma) is not always possible.

Moral luck:

Nagel’s definition of moral luck highlights that moral judgment often persists even when outcomes hinge on factors beyond control: “Where a significant aspect... depends on factors beyond his control... it can be called moral luck.” ²² No amount of technical improvement fully abolishes dependence on contingency (accidents, uncertain futures), unless one assumes an unrealizable level of control over all morally relevant outcomes.

Category mistake about legitimacy:

Habermas’s distinction (as visible in the excerpted argument) treats “instrumental action” as governed by technical rules, implying that **technical rationality is not the same as political/moral legitimacy**. ⁵ Even if a system works, the question “Who authorized it?” and “Is it just?” remains.

Technology is not just a tool (constitutive effects):

Heidegger argues the “instrumental and anthropological definition” (“means to an end”) is “correct” but not “true” in the deeper sense; focusing only on that leaves us “unfree and chained.” ⁷ Winner similarly argues technologies become “forms of life,” shaping patterns of activity and expectations; we “sleepwalk” through their world-making power. ²³ These views undermine the reduction premise (MD → TP) by claiming technology transforms the space of meaning and valuation, not merely constraints.

A clarifying map of relationships

flowchart TD

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A[Ethical conflict arises] --> B{Source of conflict?}
B -->|Scarcity / constraints| C[Expand feasible set via technology]
B -->|Empirical uncertainty| D[Improve prediction/measurement via science]
B -->|Coordination / enforcement| E[Tech-enabled institutions & artifacts]
B -->|Value conflict / legitimacy| F[Normative deliberation & authority]
C --> G[May dissolve dilemma]
D --> G
E --> G
F --> H[Not solvable by technique alone]
G --> I[New externalities & capabilities]
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I --> J[New ethical problems]
J --> B
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This diagram captures a key point: technology can transform **constraints** and **information** (C, D, E), but disputes about **ends, legitimacy, and value tradeoffs** (F) remain normative and political rather than purely technical—precisely the zone Habermas, Jonas, Sandel, Winner, and Arendt emphasize. ²⁴

Common counterarguments and pro-thesis rebuttals

Below are widely discussed objection types—and the strongest available rebuttal strategies for a defender of the thesis (not necessarily decisive).

Counterargument from “no technical solution problems” (Hardin):

Hardin claims “The population problem has no technical solution; it requires a fundamental extension in morality,” and defines “technical solution” as requiring little or no change in values. ¹

Pro-thesis rebuttal: Reject Hardin’s narrow definition: if “technical solution” includes *technologies that change incentives, identities, and values* (education tech, psychopharmacology, governance tech), then morality-change itself can be an engineered outcome. The defender can say: even “moral extension” might be technologically facilitated (though this invites deep worries about manipulation and legitimacy).

Counterargument from moral remainder (Williams):

Even correct decisions can leave “remainder.” ⁴

Pro-thesis rebuttal: Recast “remainder” as a *psychological artifact* of human bounded rationality and evolved moral emotions; sufficiently advanced cognitive enhancement (Bostrom’s emphasis on making ourselves “smarter and wiser” ¹⁷) could reduce remainder by improving consistency. This rebuttal is strongest under moral monism or consequentialism and weakest under pluralism.

Counterargument from value pluralism/incommensurability:

Pluralists argue some values are irreducibly conflicting or incommensurable. ²⁵

Pro-thesis rebuttal: Argue that any real decision procedure requires weighting; thus incommensurability is not action-guiding. Hardin himself notes “a system of weighting” is needed in practice, while conceding disagreement arises when hidden decisions become explicit. ¹ The defender can claim future tech can reveal common human values (neuroethics, preference learning) and stabilize weighting.

Counterargument from legitimacy and democracy (Habermas):

Instrumental action is technical-rule-governed, but legitimacy requires consensus and justification. ⁵

Pro-thesis rebuttal: Build “technological solutions” that are *procedural* (digital deliberation platforms, verification, transparency tools) so that technology *enables* legitimate consensus rather than replacing it. This changes the thesis from “tech decides” to “tech scaffolds decision.”

Counterargument from the non-neutrality/essence of technology (Heidegger/Winner):

Technology is not a neutral means; it shapes what counts as valuable and real. ²⁶

Pro-thesis rebuttal: Accept constitutive power and treat it as design space: if technology shapes value horizons, that is precisely the “technical” pathway by which ethics is solved—by building worlds that make better moral relations possible. This rebuttal, however, risks collapsing into authoritarian “world-building” without consent.

Empirical case studies: when tech alleviates ethical problems and when it creates new ones

The thesis is empirical as well as conceptual: technology must *in practice* deliver “solutions” across domains. The case studies below highlight both successes and recurrent failure modes.

Comparative table of case studies

Domain	Ethical problem targeted	Technological intervention	Evidence of moral benefit	New ethical problems created
Public health (vaccination)	Preventable maccob morbidity/ mortality; fairness in child survival	Routine immunization programs	WHO: immunization prevents 3.5–5M deaths/year . ³ WHO (2024): immunization efforts saved ~154M lives over 50 years. ²⁷	Distributional justice (access gaps), misinformation/ polarization, mandates vs autonomy (non-technical legitimacy disputes) ²⁸
Antibiotics / AMR	Ethical duty to treat infection vs risk of resistance	Antibiotics + stewardship/ infection control	WHO: bacterial AMR directly caused 1.27M deaths (2019) and contributed to 4.95M —showing antibiotics’ moral stakes and the cost of misuse. ²⁹	Tragedy-of-commons dynamics; intergenerational justice; global inequity—often requiring regulation and behavior change, not just new drugs ³⁰
AI in governance/ health	Improve decisions, reduce error, expand access	ML diagnostics, triage, decision support; automated systems	WHO notes AI’s “enormous potential” for health while focusing on ethical governance. ³¹ NIST AI RMF aims to manage AI risks and promote “trustworthy and responsible” use. ³²	Bias, opacity, accountability gaps, surveillance, power concentration; need for governance frameworks (NIST, UNESCO, WHO) ³³

Domain	Ethical problem targeted	Technological intervention	Evidence of moral benefit	New ethical problems created
Nuclear	Peace/energy vs catastrophic war; low-carbon power vs existential risk	Nuclear power; nuclear deterrence; safety regimes	UN: NPT aims at nonproliferation + peaceful use + disarmament. ³⁴ IAEA: safety objective is protecting people/environment from ionizing radiation. ³⁵	Weaponization and accident risk; deterrence ethics; long-lived waste; governance as essential “solution,” not engineering alone ³⁶
Bioengineering (CRISPR)	Treat genetic disease; reduce suffering	Genome editing; ex vivo therapies	FDA approved first CRISPR-based therapy (Casgevy) for sickle cell disease (12+). ³⁷ Foundational CRISPR work explicitly highlights genome-editing potential. ³⁸	Germline ethics, consent across generations, enhancement, inequality; scandals (He Jiankui) show governance failure ³⁹
Climate tech (CDR & geoengineering)	Prevent catastrophic warming; intergenerational justice	Renewables + carbon dioxide removal (CDR) + (proposed) solar geoengineering research	IPCC: CDR is “required” for net zero; cannot substitute for deep emissions cuts. ⁴⁰ National Academies recommends robust governance for solar geoengineering research. ⁴¹	Moral hazard (delay mitigation), geopolitical conflict, consent of affected populations; governance central to “solution” ⁴²

Interpreting the pattern

Across domains, technology does three consistent things:

1. **It changes what is feasible**, sometimes dissolving dilemmas that were scarcity-based (vaccines; some gene therapies). ⁴³
2. **It redistributes risk and power**, generating new ethical questions about who decides, who benefits, and who bears harm (AI governance; nuclear security). ⁴⁴

3. It creates “second-order” ethical problems (AMR as a technological commons; geoengineering governance; AI opacity). ⁴⁵

Bostrom articulates this structurally: technological revolutions “rewrite” material conditions and reshape culture, perhaps human nature. ⁸ That is a direct empirical counterweight to the thesis’s implicit assumption that technology is merely a tool we apply to a stable moral landscape.

Timeline chart: key intellectual milestones relevant to the thesis

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timeline
  title Technology-as-solution vs ethics-as-non-technical: selected milestones
  1620 : Bacon links knowledge & power over nature
  1830s : Comte positivism; science->prevision->action
  1954 : Heidegger critiques instrumental view; "essence" and enframing
  1958 : Arendt analyzes modern "world alienation" and science
  1966 : Weinberg advocates "Quick Technological Fixes" for social problems
  1968 : Hardin: "no technical solution problems"
  1973 : Jonas: technology changes action; ethics must change
  1980s : Winner: technology as politics/forms of life
  1990s : Latour: nonhumans in moral-social order (actor-network)
  2000s : Sandel: critique of enhancement/mastery
  2000s-2010s : Bostrom: technological revolutions + existential risk framing
  2020s : AI governance frameworks (NIST/UNESCO/WHO); climate tech governance (NASEM)
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Normative implications and policy recommendations

If one *accepts* the thesis, the normative implications are radical: ethics becomes a problem of R&D prioritization, systems engineering, and governance of innovation. If one *rejects* it, the implications are equally important: policymakers must treat technology as **value-laden** and insist on legitimacy and pluralism.

Normative implications

Technocratic temptation and “instrumental capture”:

Weinberg explicitly contrasts the difficulty of “social engineering” with the “simplicity” of technological engineering and asks to what extent social problems can be circumvented by converting them into technological problems. ² This supports technocratic governance styles. But Habermas’s framing warns that instrumental action’s technical-rule structure is not identical to democratic legitimacy. ⁵

Responsibility expansion and the precautionary posture:

Jonas argues modern technology changes the nature and ethical relevance of actions; older ethics assumed narrow scope of responsibility, but new power requires new ethics. ⁶ This directly motivates precaution and future-oriented governance (especially for irreversible harms).

World-making and political agency:

Arendt's emphasis on tracing "world alienation" suggests that technical progress can undermine the shared public world that makes politics possible. ¹³ Winner similarly stresses we "sleepwalk" while technologies reconstitute the conditions of existence. ¹⁵ If true, then policy must treat technology as *institution-building*, not only gadget deployment.

Policy recommendations

A graduate-level compromise stance—consistent with the evidence—is **"technological pluralism with democratic constraint"**: use technology aggressively to reduce suffering while institutionalizing safeguards that keep normative disputes visible and contestable.

Govern "solutions" as sociotechnical systems, not devices.

Adopt risk-and-rights frameworks: NIST AI RMF explicitly aims to help manage AI risks and promote "trustworthy and responsible" AI. ³² UNESCO's AI ethics recommendation centers human rights/dignity and principles like transparency, fairness, and human oversight. ⁴⁶ For health AI, WHO provides governance and ethical principles. ³¹

Require legitimacy mechanisms for high-impact tech.

Use procedural governance: transparency obligations, stakeholder deliberation, auditability. The EU explicitly frames transparency obligations and guidance processes for compliance. ⁴⁷ This is aligned with Habermas-style concerns about bringing power under citizen-consensus, even if technology supports the process. ⁵

Invest in "harm-reducing" innovation while recognizing irreducible tradeoffs.

Vaccination and gene therapy show real, measurable moral benefits from R&D. ⁴³ But AMR demonstrates that purely technical fixes are unstable without stewardship and institutions. ²⁹ Climate: IPCC states CDR is required for net zero but cannot substitute for deep emissions reductions—policy should avoid moral hazard by binding deployment to mitigation. ⁴⁰

Build governance capacity proportionate to irreversible risk.

Nuclear regimes show how governance is part of the "solution": NPT's objectives combine peaceful use, nonproliferation, and disarmament. ³⁴ IAEA's safety objective explicitly centers protection of people and environment. ³⁵ Solar geoengineering: National Academies recommends robust research governance and international mechanisms. ⁴¹

Professional ethics as a floor, not a substitute.

Codes like the ACM Code of Ethics stress that computing professionals' actions "change the world" and must support the public good. ⁴⁸ These are valuable but cannot settle contested societal values; they should be integrated into enforceable governance.

Research gaps and suggested bibliography

Research gaps

Operationalizing “technical solution” without smuggling in moral consensus:

Hardin’s definition separates technique-change from value-change, but the thesis implicitly broadens “technical” to include reshaping values or preferences—raising legitimacy concerns. ¹ More work is needed to define “technical solution” that is neither trivially broad (“everything is technical if we can engineer minds”) nor trivially narrow.

Measuring moral remainder and pluralism at population scale:

Williams’s “remainder” claim is philosophical but suggests empirical questions: how often do policy solutions leave persistent moral residue across populations? ⁴ How does technology change (or merely redistribute) felt remainder?

Value alignment as a boundary case against solutionism:

Bostrom notes enormous upside yet existential downside; success depends on “human-friendly” goal architectures. ¹⁷ This is a domain where “technical solution” and “normative specification” collide—still an open research frontier.

Global governance for techno-externalities:

AMR, climate CDR, nuclear risk, and AI all show cross-border externalities where unilateral “solutions” fail. WHO, IPCC, and NASEM emphasize governance, not only engineering. ⁴⁹ Designing legitimate international institutions remains under-developed relative to technological pace.

Suggested bibliography prioritizing originals and official sources

Primary sources (core to the thesis debate)

- Francis Bacon, *Novum Organum* (often cited for “knowledge/power” linkage). ¹¹
- Auguste Comte, *A General View of Positivism* (public-domain excerpts) and related positivist writings on law-governed society. ⁵⁰
- Martin Heidegger, “The Question Concerning Technology” (instrumental definition; essence/enframing). ⁷
- Hannah Arendt, *The Human Condition* (world alienation; Archimedean standpoint). ²⁰
- Jürgen Habermas, “Technology and Science as ‘Ideology’” / *Toward a Rational Society* (instrumental action and technical rules; politicization). ⁵
- Hans Jonas, “Technology and Responsibility: Reflections on the New Tasks of Ethics” (classic argument: technology changes action; ethics must change). ⁶
- Michael J. Sandel, “The Case Against Perfection” (mastery vs humility; “openness to the unbidden”). ¹⁴
- Langdon Winner, “Technologies as Forms of Life” (sleepwalking; progress myth; technology as world-shaping). ⁵¹
- Bruno Latour, “Technology is Society Made Durable” (delegation of morality into artifacts; humans/nonhumans). ¹⁶

- Nick Bostrom, “Superintelligence” (intelligence amplification; last invention), and “Technological Revolutions: Ethics and Policy in the Dark” (tech reshapes culture/human nature; governance difficulty). ⁵²
- Garrett Hardin, “The Tragedy of the Commons” (explicit denial of universal technical solutions; definition of technical solution). ¹
- Bernard Williams, “Ethical Consistency” (moral conflict not always soluble without remainder). ⁴
- Thomas Nagel, “Moral Luck” (definition and paradox of moral luck). ²²

Empirical and official governance sources (for case studies)

- World Health Organization: vaccines and immunization impact estimates. ⁵³
- WHO AMR fact sheet (burden estimates; drivers). ²⁹
- NIST AI RMF 1.0 (risk management; trustworthy AI). ³²
- UNESCO Recommendation on the Ethics of AI (human rights/dignity; transparency/fairness/oversight). ⁵⁴
- WHO guidance on AI for health. ³¹
- UN Office for Disarmament Affairs: NPT objectives; nuclear governance framing. ³⁴
- IAEA Fundamental Safety Principles (safety objective; radiation risk). ³⁵
- FDA press release on first CRISPR therapy approvals (Casgevy; sickle cell). ³⁷
- WHO recommendations on human genome editing governance. ⁵⁵
- IPCC AR6 WGIII CDR factsheet (CDR required for net zero; cannot substitute for deep cuts). ⁴⁰
- National Academies: *Reflecting Sunlight* (solar geoengineering research governance). ⁴¹

Bottom line: Technology very often transforms ethical landscapes and can dissolve many dilemmas by expanding feasible options, but the universal reduction “all moral dilemmas are merely technical problems” conflicts with central philosophical arguments about remainder, pluralism, legitimacy, and the constitutive nature of technology itself (Hardin, Williams, Habermas, Heidegger, Jonas, Arendt, Winner). ⁵⁶

¹ ⁵⁶ <https://math.uchicago.edu/~shmuel/Modeling/Hardin%2C%20Tragedy%20of%20the%20Commons.pdf>

<https://math.uchicago.edu/~shmuel/Modeling/Hardin%2C%20Tragedy%20of%20the%20Commons.pdf>

² https://archive.org/download/risk_management_starr_weinberg_lowrance/weinberg1966.pdf
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³ ⁴³ ⁵³ <https://www.who.int/health-topics/vaccines-and-immunization/>
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